

Name: File

Teacher/Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 2

MARCH 2006

EXTENSION 1 MATHEMATICS

Time Allowed: 70 minutes

Instructions:

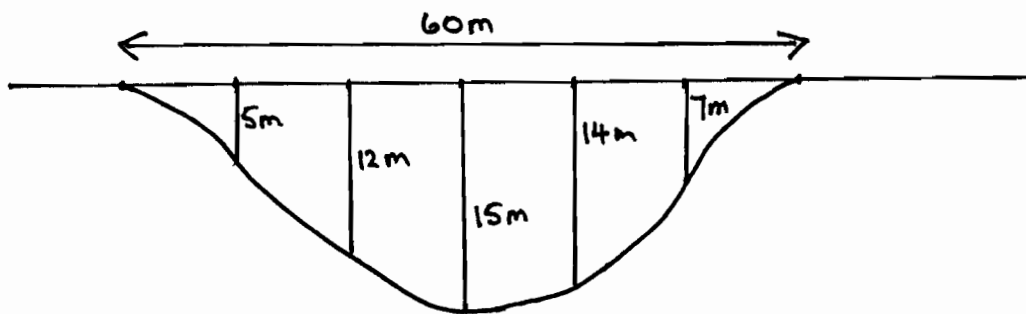
- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new page.
- Standard integrals can be found on the last page.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/10	/10	/10	/10	/10	/10	/60

Question 1 (10 marks)**Marks**

- a) Find the exact value of
- i. $\tan\left(\frac{2\pi}{3}\right)$ 1
- ii. $\sin\left(-\frac{\pi}{3}\right)$ 1
- b) Find
- $$\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$$
- 1
- c) Given that $\int_1^5 f(x) dx = 4$ find the value of k 3
- for which $\int_1^5 [f(x) + kx] dx = 28$.
- d) A river 60m wide is surveyed for its depth every 10m across its width.

The depth at each point surveyed is shown on the diagram.



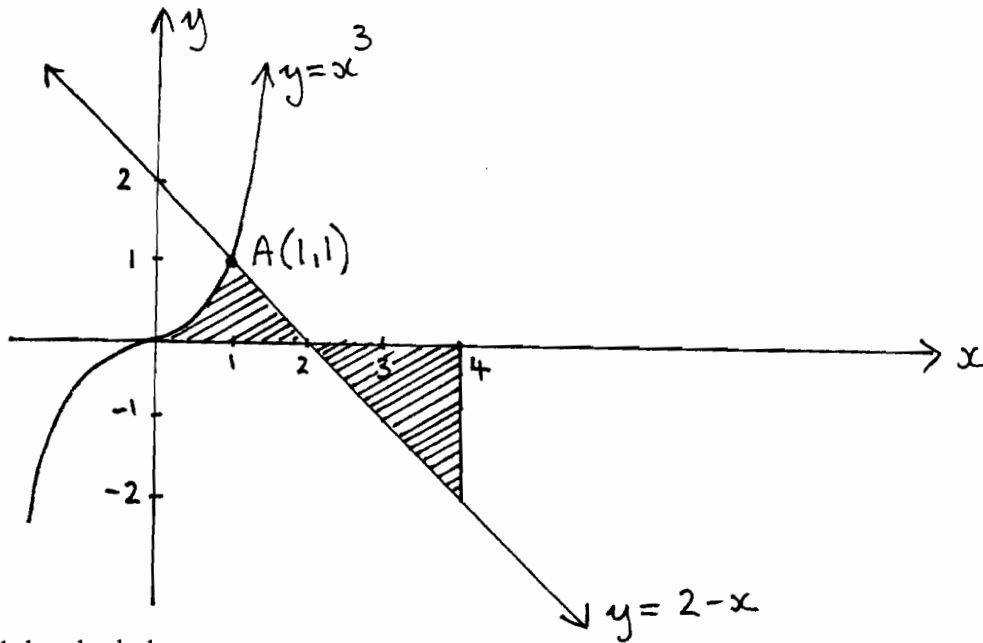
- i. Find the cross-sectional area of the river using Simpsons Rule 3
- ii. Hence find the volume of water passing this point per second if the water flows at 5m/s. 1

Question 2 (10 marks) Start a new page

Marks

- a) The point of intersection of $y = x^3$ and $y = 2 - x$ is the point A (1, 1)

3



Find the shaded area.

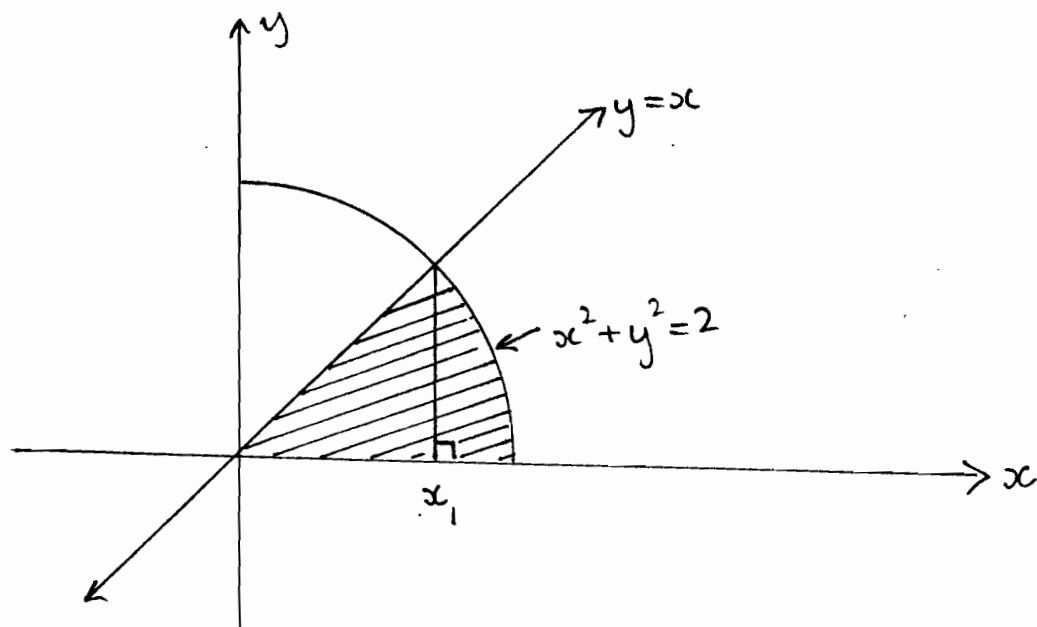
- b) If $y = \sin 2x^\circ$

- Express $2x^\circ$ in radian measure
- Find $\int \sin 2x^\circ dx$

1

2

- c)



- Find x_1
- Calculate the volume generated when the shaded region (shown above) between the line $y = x$, the circle $x^2 + y^2 = 2$ and the x axis is rotated around the x axis.

1

3

Question 3 (10 marks) Start a new page

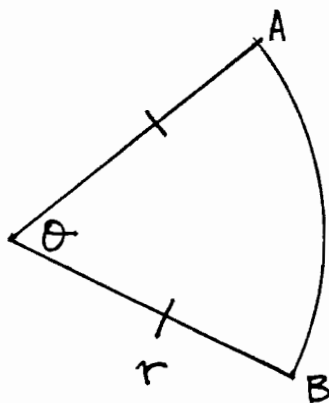
a) If $y = a \cos nx + b \sin nx$

show that $\frac{d^2 y}{dx^2} + n^2 y = 0$ 3

b) i. Differentiate $x\sqrt{x+3}$ and simplify your answer as far as possible, 2

ii. Hence find $\int \frac{x+2}{\sqrt{x+3}} dx$ 1

c) The sector below has area of 25cm^2 . It is contained in a circle of radius $r \text{ cm}$ and the arc AB subtends an angle at the centre of the circle of θ radians.



i. Show the perimeter of the sector is given by $P = 2r + \frac{50}{r}$ 1

ii. Find r for which the perimeter is a minimum. 3

Question 4 (10 marks) Start a new page

- a) i. Sketch $y = 3 \cos 2x$ for $0 \leq x \leq \pi$ 2
- ii. Find the area enclosed by $y = 3 \cos 2x$, the x axis, $x = 0$
and $x = \frac{\pi}{2}$ 3
- b) i. Express $\sin x + \sqrt{3} \cos x$ in the form $A \sin(x + \theta)$ for $0 < \theta < \frac{\pi}{2}$ 2
- ii. Hence solve $\sin x + \sqrt{3} \cos x = \sqrt{2}$ for $0 \leq x \leq 2\pi$. 3

Question 5 (10 marks) Start a new page

- a) Find $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$ using the substitution $u = \sqrt{x}$ 3
- b) Find $\int \frac{x}{\sqrt{1-x}} dx$ using the substitution $u = 1 - x$. 4
- c) Find $\int \cos^2 3x dx$ 3

Question 6 (10 marks) Start a new page

- a) Evaluate $\int_0^1 \frac{x}{(x^2 + 2)^2} dx$ using the substitution $u = x^2 + 2$ 3
- b) i. Prove $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ 1
- ii. Hence or otherwise evaluate $\int_0^{\pi/6} \sin 4x \cdot \cos 2x dx$ 3
- c) i. Sketch $y = 2^x$ 1
- ii. If n is a positive integer, by considering the graph of $y = 2^x$ 2
- show that $2^n < \int_n^{n+1} 2^x dx < 2.2^n$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$\begin{aligned} \text{a) i) } \tan\left(\frac{2\pi}{3}\right) &= \tan\left(\pi - \frac{\pi}{3}\right) \\ &= -\tan \frac{\pi}{3} \\ &= -\sqrt{3} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{ii) } \sin\left(-\frac{\pi}{3}\right) &= \sin\left(2\pi - \frac{\pi}{3}\right) \\ &= -\sin \frac{\pi}{3} \\ &= -\frac{\sqrt{3}}{2} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{b) } \lim_{x \rightarrow 0} \frac{\sin 2x}{x} &= \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \cdot 2 \\ &= 2 \end{aligned} \quad (1)$$

$$\text{c) } \int_1^5 [f(x) + kx] dx = \int_1^5 f(x) dx + \int_1^5 kx dx$$

$$4 + \left[\frac{kx^2}{2} \right]_1^5 = 28$$

$$\frac{25k}{2} - \frac{k}{2} = 24$$

$$24k = 48$$

$$k = 2$$

(3)

$$\text{d) i) } A = \frac{10}{3} (0 + 0 + 4(5 + 15 + 7) + 2(12 + 14))$$

$$A = 533 \frac{1}{3} \text{ m}^2$$

$$\begin{aligned} \text{ii) } V &= 533 \frac{1}{3} \times 5 \\ &= 2666 \frac{2}{3} \text{ m}^3 \end{aligned} \quad (1)$$

Question 2

$$\begin{aligned} \text{a) } A &= \int_0^1 x^3 dx + \frac{(1 \times 1)}{2} + \frac{(2 \times 2)}{2} \\ &= \left[\frac{x^4}{4} \right]_0^1 + 2 \frac{1}{2} \\ A &= 2 \frac{3}{4} \text{ unit}^2 \end{aligned} \quad (3)$$

$$\text{b) } \pi^\circ = 180^\circ$$

$$\begin{aligned} \text{i) } \therefore 2x^\circ &= \frac{2 \times \pi x}{180} \\ &= \frac{\pi x}{90} \text{ radians} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{ii) } \int \sin 2x^\circ dx &= \int \sin \frac{\pi}{90} x dx \\ &= -\frac{90}{\pi} \cos \frac{\pi}{90} x + C \end{aligned} \quad (2)$$

$$\begin{aligned} \text{c) i) } \text{since } y &= x \quad x^2 + y^2 = 2 \\ x^2 + x^2 &= 2 \\ 2x^2 &= 2 \end{aligned} \quad (1)$$

$$x_1 = 1$$

$$\begin{aligned} \text{ii) } V &= \pi \int_0^1 x^2 dx + \pi \int_1^{\sqrt{2}} (2 - x^2) dx \\ &= \pi \left\{ \left[\frac{x^3}{3} \right]_0^1 + \left[2x - \frac{x^3}{3} \right]_1^{\sqrt{2}} \right\} \\ &= \pi \left[\frac{1}{3} + \left(2\sqrt{2} - \frac{2\sqrt{2}}{3} \right) - \left(2 - \frac{1}{3} \right) \right] \\ &= \pi \left[-\frac{4}{3} + \frac{6\sqrt{2} - 2\sqrt{2}}{3} \right] \\ &= \pi \left[\frac{4\sqrt{2} - 4}{3} \right] \end{aligned} \quad (3)$$

Question 3

$$\text{a) } y = a \cos nx + b \sin nx$$

$$\frac{dy}{dx} = -a \sin nx + b \cos nx$$

$$\frac{d^2y}{dx^2} = -a \cos nx - b \sin nx$$

$$\text{sub into } \frac{d^2y}{dx^2} + n^2 y = 0$$

$$\begin{aligned} \text{LHS} &= -a \cos nx - b \sin nx + \\ &\quad n^2 (a \cos nx + b \sin nx) \\ &= 0 \end{aligned}$$

b) i) $y = x\sqrt{x+3}$

Let $u = x$ $v = \sqrt{x+3} = (x+3)^{1/2}$
 $u' = 1$ $v' = \frac{1}{2}(x+3)^{-1/2} = \frac{1}{2\sqrt{x+3}}$

$$\therefore \frac{dy}{dx} = \sqrt{x+3} + \frac{x}{2\sqrt{x+3}}$$

$$= \frac{2(x+3) + x}{2\sqrt{x+3}}$$

$$= \frac{3x+6}{2\sqrt{x+3}}$$

$$\frac{dy}{dx} = \frac{3}{2} \left[\frac{x+2}{\sqrt{x+3}} \right] \quad (2)$$

ii) $\therefore \int \frac{x+2}{\sqrt{x+3}} dx = \frac{2}{3} x\sqrt{x+3} + c \quad (1)$

c) i) $P = 2r + \text{arc length AB}$

$$= 2r + r\theta \quad (1)$$

since $\frac{1}{2}r^2\theta = 25$

$$\theta = \frac{50}{r^2} \text{ sub into (1)}$$

$$\therefore P = 2r + r \left[\frac{50}{r^2} \right] \quad (1)$$

$$P = 2r + \frac{50}{r} = 2r + 50r^{-1}$$

ii) $\frac{dP}{dr} = 2 - 50r^{-2}$

$$\frac{d^2P}{dr^2} = 100r^{-3}$$

stpts $2 - \frac{50}{r^2} = 0$

$$2r^2 = 50$$

$$r = \pm 5 \quad r > 0 \therefore r = 5$$

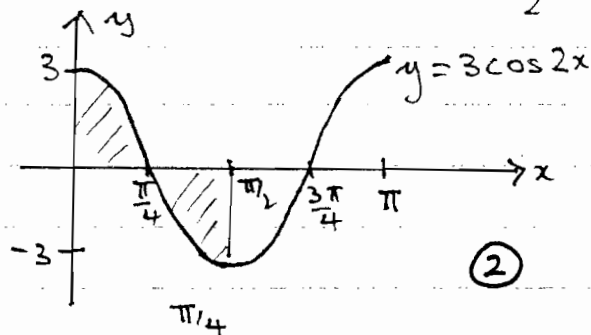
test max/min

if $r = 5$ $\frac{d^2P}{dr^2} > 0 \therefore \text{min} \quad (3)$

$\therefore$ min Perimeter if $r = 5\text{cm}$

Question 4

a) i) amplitude = 3 period $\frac{2\pi}{2} = \pi$



ii) $A = 2 \int_0^{\pi/4} 3 \cos 2x dx$

$$= 6 \left[\frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= 3 \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$= \underline{3 \text{ unit}^2} \quad (3)$$

b) i) $A = \sqrt{1+3} \therefore A = 2$

$$2 \left[\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x \right] = A \sin(x+\theta)$$

$$\cos \theta = \frac{1}{2} \quad \sin \theta = \frac{\sqrt{3}}{2} \therefore \theta = \frac{\pi}{3}$$

$$\therefore \sin x + \sqrt{3} \cos x = 2 \sin \left(x + \frac{\pi}{3} \right) \quad (2)$$

ii) $2 \sin \left(x + \frac{\pi}{3} \right) = \sqrt{2}$

$$\sin \left(x + \frac{\pi}{3} \right) = \frac{\sqrt{2}}{2}$$

$$x + \frac{\pi}{3} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \dots$$

$$\therefore x = \frac{5\pi}{12}, \frac{23\pi}{12} \quad (3)$$

Question 5

$$a) \quad u = \sqrt{x} = x^{1/2}$$

$$\frac{du}{dx} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$$

$$du = \frac{dx}{2\sqrt{x}}$$

$$\therefore dx = 2\sqrt{x} du \quad (3)$$

$$\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = \int \frac{\cos u}{\sqrt{x}} \cdot 2\sqrt{x} du$$

$$= 2 \int \cos u du$$

$$= 2 \sin u + C$$

$$= \underline{\underline{2 \sin \sqrt{x} + C}}$$

$$b) \quad u = 1-x$$

$$\frac{du}{dx} = -1$$

$$-du = dx$$

$$\int \frac{x}{\sqrt{1-x}} dx = \int \frac{1-u}{\sqrt{u}} \cdot -du$$

$$= - \int (1-u) u^{-1/2} du$$

$$= - \int (u^{-1/2} - u^{1/2}) du$$

$$= - \left[\frac{u^{1/2}}{1/2} - \frac{u^{3/2}}{3/2} \right]$$

$$(4) \quad = - \left[2\sqrt{1-x} - \frac{2}{3} \sqrt{(1-x)^3} \right] + C$$

$$c) \quad \cos 2\theta = 2\cos^2 \theta - 1$$

$$\therefore \int \cos^2 3x dx = \frac{1}{2} \int (\cos 6x + 1) dx$$

$$= \frac{1}{2} \left[\frac{1}{6} \sin 6x + x \right] + C$$

Question 6

$$a) \quad u = x^2 + 2$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

$$\int_0^1 \frac{x}{(x^2+2)^2} dx = \frac{1}{2} \int_2^3 \frac{x}{u^2} \cdot \frac{du}{x}$$

$$= \frac{1}{2} \int_2^3 u^{-2} du$$

$$= \frac{1}{2} \left[-\frac{1}{u} \right]_2^3$$

$$= \underline{\underline{\frac{1}{12}}} \quad (3)$$

$$b) i) \text{ LHS} = \sin(A+B) + \sin(A-B)$$

$$= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$$

$$= 2 \sin A \cos B$$

$$= \text{RHS} \quad (1)$$

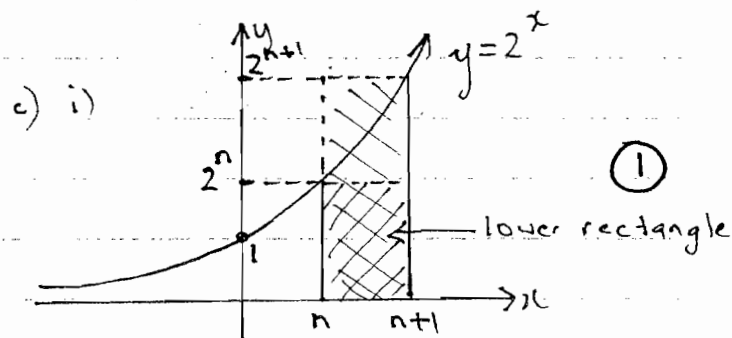
$$ii) \quad \sin 4x \cdot \cos 2x = \frac{1}{2} [\sin 6x + \sin 2x]$$

$$\frac{1}{2} \int_0^{\pi/6} (\sin 6x + \sin 2x) dx$$

$$= \frac{1}{2} \left[-\frac{1}{6} \cos 6x - \frac{1}{2} \cos 2x \right]_0^{\pi/6}$$

$$= \frac{1}{2} \left[-\frac{1}{6} \cos \pi - \frac{1}{2} \cos \frac{\pi}{3} - \left(-\frac{1}{6} - \frac{1}{2} \right) \right]$$

$$= \frac{1}{2} \left[\frac{1}{6} - \frac{1}{4} + \frac{1}{6} + \frac{1}{2} \right] = \underline{\underline{\frac{7}{24}}} \quad (3)$$



ii) area lower rectangle $< \int_n^{n+1} 2^x dx < \text{area upper rectangle}$

$$2^n \times 1 < \int_n^{n+1} 2^x dx < 2^{n+1} \times 1$$

$$2^n < \int_n^{n+1} 2^x dx < 2 \cdot 2^n$$